

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

Concept 4.1.5: CPU Performance Factors

Total Questions: 10

Mark Schemes begin on Page 6

Question 1 | Source: 9618_w25_qp_11 (Q6.b)

- (b) Increasing the number of cores in a processor can affect the performance of a computer.

Describe the drawbacks of increasing the number of cores in a processor.

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Question 2 | Source: 9618_w25_qp_12 (Q3.c)

- (c) A student buys a new computer. The table shows the specifications of the old computer and the new computer.

Old computer	New computer
1.8 GHz dual core processor	2.3 GHz dual core processor
16 MB cache	32 MB cache

Explain why increasing the clock speed **and** increasing the cache memory will improve the performance of the computer.

Clock speed

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Cache memory

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Question 3 | Source: 9618_s25_qp_13 (Q2.b)

- 2 A computer has a processor.

- (b) The computer has cache memory.

Describe **one** benefit of the computer using cache memory.

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Question 4 | Source: 9618_w24_qp_11 (Q3.b.i)

3 A computer system has a dual-core Central Processing Unit (CPU).

(b) (i) The number of cores in the processor affects the performance of the computer system.

Identify **one other** feature of a processor that can affect the performance of a computer system **and** state why it affects the performance.

Feature

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Reason

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[2]

Question 5 | Source: 9618_w23_qp_11 (Q5.c.i)

5 The Central Processing Unit (CPU) of the basic Von Neumann model for a computer system contains several special purpose registers.

(c) A computer has a single 2.1 GHz CPU.

(i) Describe how increasing the clock speed to 4 GHz can increase the performance of the computer.

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Question 6 | Source: 9618_w23_qp_11 (Q5.c.ii)

5 The Central Processing Unit (CPU) of the basic Von Neumann model for a computer system contains several special purpose registers.

(c) A computer has a single 2.1 GHz CPU.

- (ii) A second computer has a CPU with two 2.1 GHz cores.

Explain why the second computer does not always run twice as fast as the computer with one 2.1 GHz CPU.

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Question 7 | Source: 9618_s23_qp_12 (Q5.b)

- 5 A student has purchased a new laptop.

- (b) Describe **two** ways the hardware of a laptop can be upgraded to improve the performance **and** explain how each upgrade improves the performance.

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Question 8 | Source: 9618_w22_qp_11 (Q7.d)

- (d) Explain the reasons why increasing the amount of cache memory can improve the performance of a CPU.

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Question 9 | Source: 9618_w22_qp_13 (Q4.c)

- (c) Upgrading secondary storage to solid state typically improves the performance of computer systems.

Identify **one other** upgrade to the hardware **and** explain why it improves the performance of a computer system.

Upgrade

Explanation

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Question 10 | Source: 9618_s21_qp_12 (Q5.b)

5 Seth uses a computer for work.

- (b) Describe the ways in which the following factors can affect the performance of his laptop computer.

Number of cores

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Clock speed

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q6.b)

6(b)	1 mark per bullet point, max 2 marks e.g. • Latency may be increased • ... because the cores must communicate with one another • There is a potential for dead-lock situations • ... where one core may wait for information from other cores which in turn are waiting for the first one • Not all software is designed to use multi-cores • ... so some of the additional cores would be idle • Increased heat generation • ... which could cause damage to other components	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q3.c)

3(c)	1 mark per bullet point, max 2 marks for each Clock Speed: • Processor can perform more F-E cycles per second • ... so more instructions / data can be processed each second Cache Memory: • Can store more of the most frequently used instructions • ... which reduces the need to access slower RAM	4
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Mark Scheme for Question 3 | Source: 9618_s25_ms_13 (Q2.b)

2(b)	1 mark each • Using cache memory improves system performance • because cache is fast access memory close to the CPU • which stores frequently used instructions / data • so that they can be accessed faster than from RAM	2
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Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q3.b.i)

3(b)(i)	1 mark for the feature, 1 mark max for the matching reason e.g. Feature: clock speed - Higher clock speed means that more F-E cycles are executed per second // Higher clock speed results in more throughput Feature: bus width - Larger bus width means that more data transferred at the same time	2
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Mark Scheme for Question 5 | Source: 9618_w23_ms_11 (Q5.c.i)

5(c)(i)	1 mark for each bullet point (max 1) - The CPU can now perform nearly twice as many F-E cycles per second - Instead of 2.1 billion F-E cycles per second, the CPU can now perform 4 billion FE cycles per second	1
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Mark Scheme for Question 6 | Source: 9618_w23_ms_11 (Q5.c.ii)

5(c)(ii)	1 mark for each bullet point (max 5) - Multiple cores introduce additional overheads ...because of the need for communication between cores - Software may not be designed for multiple cores... ...so one of the cores will be left idle - Memory access speed may not match speed of cores... ...so causing delay - The two computers may have more differences than just the cores ...one may have more RAM which allows faster multitasking ...one may have a GPU ...etc.	5
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Mark Scheme for Question 7 | Source: 9618_s23_ms_12 (Q5.b)

5(b)	1 mark for description; 1 mark for corresponding explanation Examples • Increase number of cores • Each core can independently carry out a process at the same time // so that more instructions are performed in parallel • Increase RAM capacity • ... allowing more applications to reside in memory at the same time, saving disk access times • Increase cache memory • More data can be stored in fast access so less time is spent accessing from RAM • Increase clock speed • More Fetch-Decode-Execute (FDE) cycles can run each second / per unit time	4
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Mark Scheme for Question 8 | Source: 9618_w22_ms_11 (Q7.d)

7(d)	1 mark for each bullet point (max 2): • cache is fast access memory (close to the CPU) • cache stores frequently used instructions / data • ... more cache means more instructions / data can be transferred faster • ... less swapping between RAM and cache • prevents the CPU idling while waiting for data	2
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Mark Scheme for Question 9 | Source: 9618_w22_ms_13 (Q4.c)

4(c)	1 mark for identification of a correct upgrade: 1 mark for a corresponding explanation: Examples: • increase quantity of RAM • ... so allowing more applications to reside in memory at the same time, saving disk access times • increase the size of cache memory • ... so that the CPU can continue working without waiting for data • increase clock speed • ... so that more instructions are performed in a time period • increase the number of processors / cores • ... so that more instructions are performed in parallel	2
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Mark Scheme for Question 10 | Source: 9618_s21_ms_12 (Q5.b)

5(b)	1 mark per bullet point to max 3 per factor. max 4 overall. Number of cores: • Each core processes one <u>instruction</u> per clock pulse • More/multiple cores mean that sequences of instructions can be split between them • ... and so more than one <u>instruction</u> is executed per clock pulse // more sequences of instructions can be run at the same time • More cores decreases the time taken to complete task Clock speed: • Each <u>instruction</u> is executed on a clock pulse // one F-E cycle is run on each clock pulse • ... so the clock speed dictates the number of <u>instructions</u> that can be run per second • The faster the clock speed the more <u>instructions</u> can be run per second	4
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